

MODULE II CONSTRAINTS AND SQL COMMANDS

Course Code: 23IT201

Course Title: DataBase Management Systems

Academic Year: 2024 - 2025 (Even Sem)

MODULE II CONSTRAINTS AND SQL COMMANDS

DDL Commands - Create, Drop, Alter, Truncate, Rename,

Normalization - 1NF, 2NF, 3NF, BCNF, 4NF-

DML Commands - Select, Insert, Update, Delete, Any, All, In, Exists, Non Exists,
Union, Intersection,

Subqueries - nested, correlated, Joins- Inner, Outer, and Equi, Functions - SUM,
COUNT, AVG, MIN,

MAX, Clauses - Group By, Having By, Embedded SQL, Dynamic SQL,

Transaction Concepts –

Transaction model – ACID Properties – Serializability

SQL Constructs-DDL,DML,TCL,DCL

Topics to be Covered

- Data Definition Language (DDL)
- Data Manipulation Language (DML)
- Transaction Control Language (TCL)
- Data Control Language (DCL)
- Data Manipulation Operations – Additional Basic Operations
- Data Manipulation Operations – Set Operations

Data Definition Language (DDL)

- Component of Structured Query Language (SQL- Sequel)
- Used for specifying database schema
- Used to create and modify the structure of database objects in a database
- Also known as Data Description Language
- SQL – Case In-sensitive (case doesn't matter. Eg., CREATE or create or cReatE)



DDL:



Data Definition Language (DDL)

Commands in DDL

- CREATE – To create database instance
- DROP – To drop database instances
- ALTER – To alter structure of database
- RENAME – To rename database instances
- TRUNCATE – To delete tables in a database instance

Naming Conventions For Table Name

- ✓ First letter should be alphabet and Reserve word cannot be used
- ✓ Maximum length for table name is 30 characters
- ✓ Two different tables cannot have same name
- ✓ Underscore, numerals & letters are allowed
- ✓ Blank space and Single quotes are not allowed

Data Definition Language (DDL)

Create

- Creates database instance (object)

Data Type

- Type of data to be stored in a field. E.g., int, float, date, time, char, varchar

Size

- Total length of a data object in a field

Syntax

CREATE TABLE table_name(column1_name data_type(size),column2_name data_type(size),.....);

E.g., CREATE TABLE students(id int,first_name varchar(15),last_name varchar(15),phone_no int,experience int);

Output



Table created.

Data Definition Language (DDL)

Desc

- To view the structure of the table

Syntax

DESC table_name;

E.g., DESC students;

Output



TABLE STUDENTS

Column	Null?	Type
ID	-	NUMBER
FIRST_NAME	-	VARCHAR2(15)
LAST_NAME	-	VARCHAR2(15)
PHONE_NO	-	NUMBER
EXPERIENCE	-	NUMBER

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5 rows selected.

Data Definition Language (DDL)

Drop

- Delete a database instance (object)
- Used to remove a relation (base table) and its definition
- The relation can no longer be used in queries, updates, or any other commands since its description no longer exists

Syntax

DROP TABLE table_name;

E.g., DROP TABLE students;

Output

DESC students;



Table dropped.

ORA-20001: object STUDENTS does not exist

Data Definition Language (DDL)

Alter

- Modify the structure of a table or database
- ✓ Add new columns to a table
- ✓ Change the data type or size of a column

Syntax

ALTER TABLE table_name MODIFY(column_name newdata_type(newsize));
E.g., ALTER TABLE students MODIFY (first_name varchar(20));

Output

DESC students;



Table altered.

TABLE STUDENTS

Column	Null?	Type
ID	-	NUMBER
FIRST_NAME	-	VARCHAR2(20)
LAST_NAME	-	VARCHAR2(15)
PHONE_NO	-	NUMBER
EXPERIENCE	-	NUMBER

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Data Definition Language (DDL)

Alter

Syntax

ALTER TABLE table_name ADD(column_name data_type(size));

E.g., ALTER TABLE students ADD(gender varchar(5));

Output



Table altered.

DESC students;

Restrictions on the Alter Table

- Alter table cannot be used
 - ✓ to change name of the table
 - ✓ to change name of the column
 - ✓ to drop a column
 - ✓ to decrease size of a column if table data exists

TABLE STUDENTS

Column	Null?	Type
ID	-	NUMBER
FIRST_NAME	-	VARCHAR2(20)
LAST_NAME	-	VARCHAR2(15)
PHONE_NO	-	NUMBER
EXPERIENCE	-	NUMBER
GENDER	-	VARCHAR2(5)

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6 rows selected.

Data Definition Language (DDL)

Rename

- To change the name of a table in a database

Syntax

RENAME oldtable_name TO newtable_name;

E.g., RENAME students TO employees;

Output



statement processed.
... employees;

TABLE EMPLOYEES		
Column	Null?	Type
ID	-	NUMBER
FIRST_NAME	-	VARCHAR2(20)
LAST_NAME	-	VARCHAR2(15)
PHONE_NO	-	NUMBER
EXPERIENCE	-	NUMBER
GENDER	-	VARCHAR2(5)

Download CSV
6 rows selected.

Data Definition Language (DDL)

Truncate

- To delete all the data in a table without affecting table structure

Syntax

TRUNCATE TABLE table_name;

E.g., TRUNCATE TABLE employees;

Output



INSERT INTO employees VALUES('1','sindhu','karthi','9944714665','4','F');

1 row(s) inserted.

ROM employees;

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE	GENDER
1	sindhu	karthi	9944714665	4	F

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Data Definition Language (DDL)

Truncate

Output



TRUNCATE TABLE employees;

Table truncated.

SELECT * FROM employees;

no data found

SELECT * FROM employees;

TABLE EMPLOYEES

Column	Null?	Type
ID	-	NUMBER
FIRST_NAME	-	VARCHAR2(20)
LAST_NAME	-	VARCHAR2(15)
PHONE_NO	-	NUMBER
EXPERIENCE	-	NUMBER
GENDER	-	VARCHAR2(5)

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6 rows selected.

Data Definition Language (DDL)

Drop vs Truncate

- The Difference between Drop and Truncate is as follows:

DROP	TRUNCATE
SQL command that destroys the table structure and the data stored in it	SQL command that helps to remove the records of a table
Helps to remove the records of the table, table structure and to remove the database from the system	Helps to remove the records from the table
Drop table table_name;	Truncate table table_name;

Data Manipulation Language (DML)

- Component of Structured Query Language (SQL- Sequel)
 - Used for accessing and manipulating data in a database
 - Used to add, delete or modify an existing data in a table
 - Add a new data in a table
 - Change the data of a record in a table
 - Delete the data of a record in table
 - DML is of two types
- ✓ Procedural deals with “what” and “how”. E.g., Relational Algebra
 - ✓ Non-Procedural deals with “what” only. E.g., Relational Calculus

DML:



SELECT

INSERT

UPDATE

DELETE

Data Manipulation Language (DML)

Commands in DML

- SELECT – To read records from table(s)
- INSERT – To insert record(s) into the table(s)
- UPDATE – Update the data in table(s)
- DELETE – Delete all the records from the tables

Select

- Used to select a set of data or information from a table
- Retrieve information or view a set of information
- Used to query information from a table
- Also known as Data Query Language (DQL)
- Select command is used with different conditions and clauses

Data Manipulation Language (DML)

Select

- Clauses used are Select, From and Where

Note: The table should not be empty (If the table is empty, it will give an error)

Select Clause & From Clause

- Select clause lists the attributes desired in the result of a query

Syntax 1

SELECT column1_name,column2_name,..... FROM table_name;

E.g., SELECT id,first_name,last_name FROM students;

Output



ID	FIRST_NAME	LAST_NAME
1	sindhu	karthi

[Download CSV](#)

Data Manipulation Language (DML)

Select Clause & From Clause

- From clause lists the relations involved in the query

Syntax 2

SELECT * FROM table_name;

E.g., SELECT * FROM students;

Output



ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE	GENDER
1	sindhu	karthi	9944714665	4	F

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Data Manipulation Language (DML)

Where Clause

- Where clause specifies conditions that the result must satisfy

Syntax 1

SELECT column_name FROM table_name WHERE condition;

E.g., SELECT id FROM students WHERE experience=2;

Output



Before

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE
1	sindhu	karthi	9944714665	4
2	sindhu	ranjith	9984774665	5
3	sindhu	karthi	8984774703	2
1	sindhu	karthi	9944714665	4

Download CSV
4 rows selected.

After

ID
3

Download CSV

Data Manipulation Language (DML)

Where Clause

Syntax 2

SELECT * FROM table_name WHERE condition;

E.g., SELECT * FROM students WHERE experience=2;

Output



Before

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE
1	sindhu	karthi	9944714665	4
2	sindhu	ranjith	9984774665	5
3	sindhu	karthi	8984774703	2
1	sindhu	karthi	9944714665	4

Download CSV
4 rows selected.

After

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE
3	sindhu	karthi	8984774703	2

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Data Manipulation Language (DML)

Insert

- Used to add new records or rows to a table

Syntax 1

```
INSERT INTO table_name(column1_name,column2_name,.....)
VALUES(value1,value2,.....);
```

Note: Order of the columns in the table does not matter. We can write it any ways

```
E.g., INSERT INTO students(id,first_name,last_name,phone_no,experience)
VALUES('1','sindhu','karthi','9944714665','4');
```

Output



1 row(s) inserted. ROM students;

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE
1	sindhu	karthi	9944714665	4

[Download CSV](#)

Data Manipulation Language (DML)

Insert

Syntax 2

INSERT INTO table_name VALUES(value1,value2,.....);

Note: Values are automatically added to the columns in the table. Order of the columns in the table matters. We have to remember the order of the table.

E.g., INSERT INTO students VALUES('1','sindhu','karthi','9944714665','4');

Output



1 row(s) inserted.
ROM students;

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE
1	sindhu	karthi	9944714665	4
2	sandhya	ranjith	9944714665	5

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Data Manipulation Language (DML)

Update

- Used to change or modify data in a table

Note: WHERE clause is very important

Syntax

UPDATE table_name SET column_name1=value1,column2_name2=value2,
..... WHERE condition;

E.g., UPDATE students SET phone_no=8015491040 WHERE id=2;

Output



1 row(s) updated.

FROM students;

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE
1	sindhu	karthi	9944714665	4
2	sandhya	ranjith	8015491040	5

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Data Manipulation Language (DML)

Delete

- Used to delete a record or a data from a table

Note: WHERE clause is very important. If WHERE clause is not included, all the data in the table will be deleted

Syntax

DELETE FROM table_name WHERE condition;

E.g., DELETE FROM students WHERE id=2;

Output



1 row(s) deleted.
SELECT FROM students;

ID	FIRST_NAME	LAST_NAME	PHONE_NO	EXPERIENCE
1	sindhu	karthi	9944714665	4

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Transaction Control Language (TCL)

- Component of SQL used to manage transactions in a database
- Used to manage changes made to the data in a table by DML statements
- Also allows statements to be grouped together into logical transactions
- TCL control transactions in a table
- Transactions are any operation performed on a table using DML

COMMIT

ROLLBACK

SAVEPOINT

TCL:



Transaction Control Language (TCL)

Rollback

- Used to get back to the previous permanent status of the table
- Database state is restored to what it was before the first statement of the transaction was executed
- Similar to undo command
- Note: Table can be rolled-back only if it is temporary. If committed the changes, it cannot be rolled-back

Syntax

ROLLBACK TO savepoint_name;

Transaction Control Language (TCL)

Savepoint

- Used along with the rollback command to mark a transaction in a table
- A transaction can be named using this command
- Similar to bookmarks

Syntax

SAVEPOINT savepoint_name;

Commit

- Used to save the changes made to the table permanently
- After the transaction is committed, a new transaction is automatically started

Syntax

COMMIT;

Transaction Control Language (TCL)

E.g.,

```
INSERT INTO class VALUES(5, 'Rahul');  
COMMIT;  
UPDATE class SET name = 'Abhijit' WHERE id = '5';  
SAVEPOINT A;  
INSERT INTO class VALUES(6, 'Chris');  
SAVEPOINT B;  
INSERT INTO class VALUES(7, 'Bravo');  
SAVEPOINT C;  
SELECT * FROM class;
```

Output

class	
id	name
1	Abhi
2	Adam
4	Alex

id	name
1	Abhi
2	Adam
4	Alex
5	Abhijit
6	Chris
7	Bravo

Transaction Control Language (TCL)

E.g.,
ROLLBACK TO B;
SELECT * FROM class;

ROLLBACK TO A;
SELECT * FROM class;

Output

id	name
1	Abhi
2	Adam
4	Alex
5	Abhijit
6	Chris

id	name
1	Abhi
2	Adam
4	Alex
5	Abhijit

Data Control Language (DCL)

- Used to control user access to data stored in a database
- Component of SQL which is related to the security issues of a database
- Privileges are of two types
- ✓ **System:** Includes permissions for creating session, table, etc and all types of other system privileges
- ✓ **Object:** Includes permissions for any command or query to perform any operation on the database tables
- Commands of DCL
- ✓ **Grant:** Used to give permission to a user to access a database
- ✓ **Revoke:** Used to withdraw permission from a user to access a database

DCL:



GRANT

REVOKE

Data Control Language (DCL)

- **Allow a User to create session:** To grant the session creating privileges

Syntax

GRANT CREATE SESSION TO username;

- **Allow a User to create table:** To allow a user to create tables in the database

Syntax

GRANT CREATE TABLE TO username;

- **Grant all privilege to a User:** sysdba is a set of privileges which has all the permissions in it

Syntax

GRANT sysdba TO username;

Data Control Language (DCL)

- **Provide user with space on tablespace to store table**
- ✓ Allowing a user to create table is not enough to start storing data in that table
- ✓ Provide user with privileges to use available tablespace for their table & data

Syntax

ALTER USER username QUOTA UNLIMITED ON SYSTEM;

- ✓ The above command will alter the user details and will provide it access to unlimited tablespace on system
- ✓ **Note:** Generally unlimited quota is provided to Admin users
- To prevent a user from creating table in a database

Syntax

REVOKE CREATE ANY TABLE TO username;

Data Control Language (DCL)

- **Grant permission to create any table:** Grant privileges to a user to create any table

Syntax

GRANT CREATE ANY TABLE TO username;

- **Grant permission to drop any table:** Allow user to drop any table from the database

Syntax

GRANT DROP ANY TABLE TO username;

- **To take back Permissions:** To take back the privileges from any user

Syntax

REVOKE CREATE TABLE FROM username;

Data Manipulation Operations – Additional Basic Operations

The Rename Operation

- Uses the as clause which can appear in both the select and from clauses

Syntax

old-name AS new-name

E.g. 1, “For renaming the attribute name name with the name instructor name”

```
SELECT name AS instructor name, course id FROM instructor, teaches  
WHERE instructor.ID= teaches.ID;
```

E.g. 2, “For all instructors in the university who have taught some course, find their names and the course ID of all courses they taught”

```
SELECT T.name, S.course id FROM instructor AS T, teaches AS S where T.ID=  
S.ID;
```

Data Manipulation Operations – Additional Basic Operations

The Rename Operation

E.g. 3, “Find the names of all instructors whose salary is greater than at least one instructor in the Biology department”

```
SELECT DISTINCT T.name FROM instructor AS T, instructor AS S WHERE  
T.salary > S.salary AND S.dept name = 'Biology';
```

Attribute Specification in Select Clause

- Asterisk symbol “ * ” can be used in the select clause to denote “all attributes”

E.g., `SELECT instructor.* FROM instructor, teaches WHERE instructor.ID=teaches.ID;`

Data Manipulation Operations – Additional Basic Operations

Where Clause Predicates

- Includes a between comparison operator to simplify where clauses that specify that a value be less than or equal to some value and greater than or equal to some other value

E.g. , “To find the names of instructors with salary amounts between \$90,000 and \$100,000”

```
SELECT name FROM instructor WHERE salary BETWEEN 90000 AND 100000;
```

Data Manipulation Operations – Additional Basic Operations

String Operations

- SQL specifies strings by enclosing them in single quotes E.g., 'Computer'
- Single quote character that is part of a string can be specified by using two single quote characters E.g., "It's right" can be specified by "It''s right"
- SQL also permits a variety of functions on character strings
 - ✓ concatenating (using "||")
 - ✓ extracting substrings
 - ✓ finding the length of strings
 - ✓ converting strings to uppercase (using the function upper(s) where s is a string) and lowercase (using the function lower(s))
 - ✓ removing spaces at the end of the string (using trim(s)) and so on

Data Manipulation Operations – Additional Basic Operations

String Operations

- Pattern matching can be performed on strings, using the operator like
- Describe patterns by using two special characters
- ✓ Percent (%): The % character matches any substring
- ✓ Underscore (_): The character matches any character
- Patterns are case sensitive E.g., 'Intro%' matches any string beginning with "Intro"
- ✓ '%Comp%' matches any string containing "Comp" as a substring E.g., 'Intro. to Computer Science', and 'Computational Biology'
- ✓ '___' matches any string of exactly three characters
- ✓ '___%' matches any string of at least three characters

Data Manipulation Operations – Additional Basic Operations

String Operations

E.g., `SELECT dept name FROM department WHERE building LIKE '%Watson%'`

- The escape character (`\`) is used immediately before a special pattern character to indicate that the special pattern character is to be treated like a normal character

E.g. 1, like `'ab\%cd%'` escape `'\'` matches all strings beginning with “ab%cd”

E.g. 2, like `'ab\\cd%'` escape `'\'` matches all strings beginning with “ab\cd”

Data Manipulation Operations – Additional Basic Operations

The Order By Clause

- Offers the user some control over the order to display tuples in a relation
- Order by clause arranges the tuples in the result of a query in sorted order

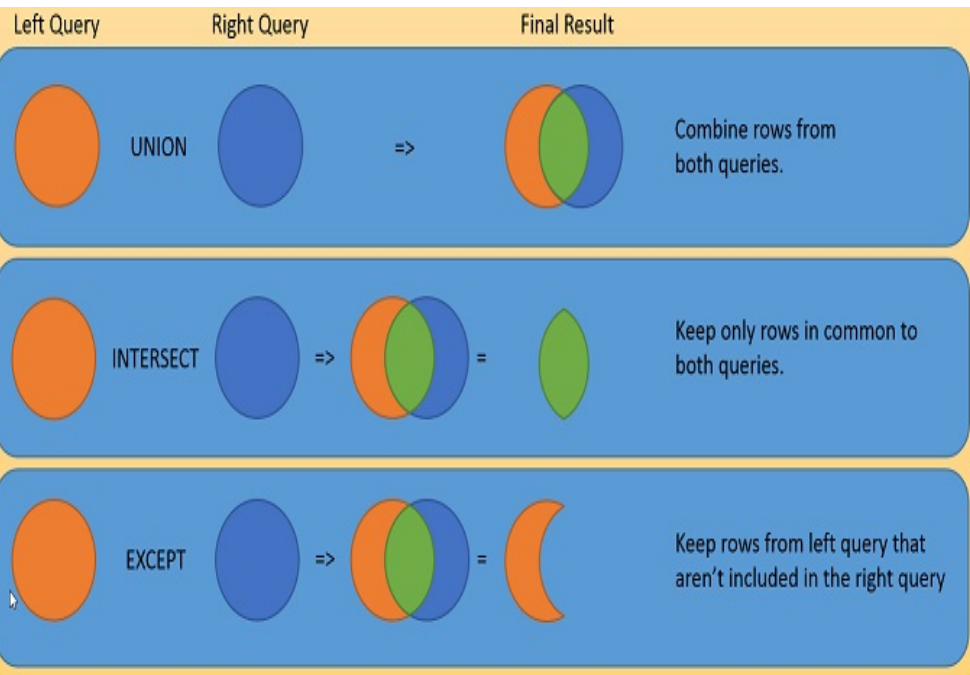
E.g. 1, “To list in alphabetic order all instructors in the Physics department”

```
SELECT name FROM instructor WHERE dept name = 'Physics' ORDER BY  
name;
```

E.g. 2, “To list the entire instructor relation in descending order of salary, if several instructors have the same salary, then order them in ascending order by name”

```
SELECT * FROM instructor ORDER BY salary DESC, name ASC;
```

Data Manipulation Operations – Set Operations



course_id	sec_id	semester	year	building	room_number	time_slot_id
BIO-101	1	Summer	2009	Painter	514	B
BIO-301	1	Summer	2010	Painter	514	A
CS-101	1	Fall	2009	Packard	101	H
CS-101	1	Spring	2010	Packard	101	F
CS-190	1	Spring	2009	Taylor	3128	E
CS-190	2	Spring	2009	Taylor	3128	A
CS-315	1	Spring	2010	Watson	120	D
CS-319	1	Spring	2010	Watson	100	B
CS-319	2	Spring	2010	Taylor	3128	C
CS-347	1	Fall	2009	Taylor	3128	A
EE-181	1	Spring	2009	Taylor	3128	C
FIN-201	1	Spring	2010	Packard	101	B
HIS-351	1	Spring	2010	Painter	514	C
MU-199	1	Spring	2010	Packard	101	D
PHY-101	1	Fall	2009	Watson	100	A

The section relation.

- Union, Intersect, and Except operate on relations and correspond to the mathematical set-theory operations $\cup$, $\cap$, and $-$ over two sets

Set Operations: Part 1:

Part 2:

Part 3:

Data Manipulation Operations – Set Operations

The Union Operation ($\cup$)

- Automatically eliminates duplicates, unlike the select clause

E.g., “To find the set of all courses taught either in Fall 2009 or in Spring 2010, or both”

(SELECT course id FROM section WHERE semester = 'Fall' AND year= 2009) UNION (SELECT course id FROM section WHERE semester = 'Spring' AND year= 2010);

- To retain all duplicates, use union all in place of union

E.g., (SELECT course id FROM section WHERE semester = 'Fall' AND year= 2009) UNION ALL (SELECT course id FROM section WHERE semester = 'Spring' AND year= 2010);

Output

course_id
CS-101
CS-315
CS-319
CS-347
FIN-201
HIS-351
MU-199
PHY-101

Data Manipulation Operations – Set Operations

The Intersect Operation ($\cap$)

- “To find the set of all courses taught in the Fall 2009 as well as in Spring 2010”

E.g., (SELECT course id FROM section WHERE semester = ‘Fall’ AND year= 2009) INTERSECT (SELECT course id FROM section WHERE semester = ‘Spring’ AND year= 2010);

Output

course_id
CS-101

- To retain all duplicates, use intersect all in place of intersect

E.g., (SELECT course id FROM section WHERE semester = ‘Fall’ AND year= 2009) INTERSECT ALL (SELECT course id FROM section WHERE semester = ‘Spring’ AND year= 2010);

Data Manipulation Operations – Set Operations

The Except Operation (-)

- Performs set difference
- “To find all courses taught in the Fall 2009 semester but not in the Spring 2010 semester”

E.g., (SELECT course id FROM section WHERE semester = 'Fall' AND year= 2009) EXCEPT (SELECT course id FROM section WHERE semester = 'Spring' AND year= 2010);

- To retain duplicates, use except all in place of except

E.g., (SELECT course id FROM section WHERE semester = 'Fall' AND year= 2009) EXCEPT ALL (SELECT course id FROM section WHERE semester = 'Spring' AND year= 2010);

Output

course_id
CS-347
PHY-101

Data Manipulation Operations – Set Operations

Cartesian Product

- Combines tuples from two tables that are unrelated to each other
- Each tuple in first table is combined with every tuple in second table

E.g., `SELECT * FROM instructor, teaches WHERE instructor.ID= teaches.ID;`

Output

ID	name	dept_name	salary
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

The instructor relation.

ID	course_id	sec_id	semester	year
10101	CS-101	1	Fall	2009
10101	CS-315	1	Spring	2010
10101	CS-347	1	Fall	2009
12121	FIN-201	1	Spring	2010
15151	MU-199	1	Spring	2010
22222	PHY-101	1	Fall	2009
32343	HIS-351	1	Spring	2010
45565	CS-101	1	Spring	2010
45565	CS-319	1	Spring	2010
76766	BIO-101	1	Summer	2009
76766	BIO-301	1	Summer	2010
83821	CS-190	1	Spring	2009
83821	CS-190	2	Spring	2009
83821	CS-319	2	Spring	2010
98345	EE-181	1	Spring	2009

The teaches relation.

inst.ID	name	dept_name	salary	teaches.ID	course_id	sec_id	semester	year
10101	Srinivasan	Physics	95000	10101	CS-101	1	Fall	2009
10101	Srinivasan	Physics	95000	10101	CS-315	1	Spring	2010
10101	Srinivasan	Physics	95000	10101	CS-347	1	Fall	2009
10101	Srinivasan	Physics	95000	10101	FIN-201	1	Spring	2010
10101	Srinivasan	Physics	95000	15151	MU-199	1	Spring	2010
10101	Srinivasan	Physics	95000	22222	PHY-101	1	Fall	2009
...	...	...	...	...	...	...	...	...
12121	Wu	Physics	95000	10101	CS-101	1	Fall	2009
12121	Wu	Physics	95000	10101	CS-315	1	Spring	2010
12121	Wu	Physics	95000	10101	CS-347	1	Fall	2009
12121	Wu	Physics	95000	10101	FIN-201	1	Spring	2010
12121	Wu	Physics	95000	15151	MU-199	1	Spring	2010
12121	Wu	Physics	95000	22222	PHY-101	1	Fall	2009
...	...	...	...	...	...	...	...	...
15151	Mozart	Physics	95000	10101	CS-101	1	Fall	2009
15151	Mozart	Physics	95000	10101	CS-315	1	Spring	2010
15151	Mozart	Physics	95000	10101	CS-347	1	Fall	2009
15151	Mozart	Physics	95000	10101	FIN-201	1	Spring	2010
15151	Mozart	Physics	95000	15151	MU-199	1	Spring	2010
15151	Mozart	Physics	95000	22222	PHY-101	1	Fall	2009
...	...	...	...	...	...	...	...	...
22222	Einstein	Physics	95000	10101	CS-101	1	Fall	2009
22222	Einstein	Physics	95000	10101	CS-315	1	Spring	2010
22222	Einstein	Physics	95000	10101	CS-347	1	Fall	2009
22222	Einstein	Physics	95000	10101	FIN-201	1	Spring	2010
22222	Einstein	Physics	95000	15151	MU-199	1	Spring	2010
22222	Einstein	Physics	95000	22222	PHY-101	1	Fall	2009
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...

The Cartesian product of the instructor relation with the teaches relation.

Data Manipulation Operations – Set Operations

- Present special problems in relational operations, including arithmetic operations, comparison operations, and set operations
- The result of an arithmetic expression (involving, for example +, −, *, or /) is null if any of the input values is null
- Comparisons involving nulls are more of a problem
- “To find all instructors who appear in the instructor relation with null values for salary”

E.g., `SELECT name FROM instructor WHERE salary IS NULL;`

Data Manipulation Operations – Set Operations

- Functions that take a collection (a set or multiset) of values as input and return a single value
 - ✓ Average: avg
 - ✓ Minimum: min
 - ✓ Maximum: max
 - ✓ Total: sum
 - ✓ Count: count
 - ✓ “Find the average salary of instructors in the Computer Science department”

E.g., `SELECT AVG (salary) FROM instructor WHERE dept name= 'Comp. Sci.';`

Aggregate Functions:



Data Manipulation Operations – Set Operations

The Group By Clause

- To apply the aggregate function to a single or a group of sets of tuples
- “To find the average salary in each department”

E.g., `SELECT dept name, AVG (salary) AS avg salary FROM instructor
GROUP BY dept name;`

The Having Clause

- Useful to state a condition that applies to groups rather than to tuples
- “To find only the departments where the average salary of the instructors is more than \$42,000”

E.g., `SELECT dept name, AVG (salary) AS avg salary FROM instructor
GROUP BY dept name HAVING AVG (salary) > 42000;`

Data Manipulation Operations – Set Operations

- A subquery is a select-from-where expression that is nested within another query
- in and not in operators can also be used on enumerated sets

Subqueries in the Where Clause

- “To find all the courses taught in the Fall 2009 semester and also in the Spring 2010 semester”

E.g. 1, `SELECT DISTINCT course id FROM section WHERE semester = 'Fall' AND year= 2009 AND course id IN (SELECT course id FROM section WHERE semester = 'Spring' AND year= 2010);`

- “To find all the courses taught in the Fall 2009 semester but not in the Spring 2010 semester”

Data Manipulation Operations – Set Operations

Subqueries in the Where Clause

E.g. 2, `SELECT DISTINCT course id FROM section WHERE semester = 'Fall' AND year= 2009 AND course id NOT IN (SELECT course id FROM section WHERE semester = 'Spring' AND year= 2010);`

Subqueries in the From Clause

- “To find the average instructors’ salaries of those departments where the average salary is greater than \$42,000” without using the having clause

E.g., `SELECT dept name, AVG salary FROM (SELECT dept name, AVG (salary) AS avg_salary FROM instructor GROUP BY dept name) WHERE avg_salary > 42000;`

Data Manipulation Operations – Set Operations

The with Clause

- Provides a way of defining a temporary relation whose definition is available only to the query in which the with clause occurs
- “To finds those departments with the maximum budget”

E.g., WITH max_budget (value) AS (SELECT max(budget) FROM department)
SELECT budget FROM department, max_budget WHERE department.budget
= max_budget.value;

Nested Subqueries:



SQL Constructs-DDL,DML,TCL,DCL

Summary

- Data Definition Language (DDL)
- Data Manipulation Language (DML)
- Transaction Control Language (TCL)
- Data Control Language (DCL)
- Data Manipulation Operations – Additional Basic Operations
- Data Manipulation Operations – Set Operations

SQL Constructs-DDL,DML,TCL,DCL

Try Yourself

- Use DDL commands to create and modify the following tables
 1. Person (driverid, name, address, phoneno, age)
 2. Car (license, model, year)
 3. Accident (reportno, date, location)
 4. Product (manufacturer, modelno, type)
 5. Laptop (modelno, speed, ram size, hard disk size, screen, price)
 6. Book (title, author, publisher, category, year, price)
 7. Distributor (distid, distname, city, discount)

SQL Constructs-DDL,DML,TCL,DCL

Try Yourself

- Use DML commands to add, delete or modify the data in the following tables
 1. Person (driverid, name, address, phoneno, age)
 2. Car (license, model, year)
 3. Accident (reportno, date, location)
 4. Product (manufacturer, modelno, type)
 5. Laptop (modelno, speed, ram size, hard disk size, screen, price)
 6. Book (title, author, publisher, category, year, price)
 7. Distributor (distid, distname, city, discount)